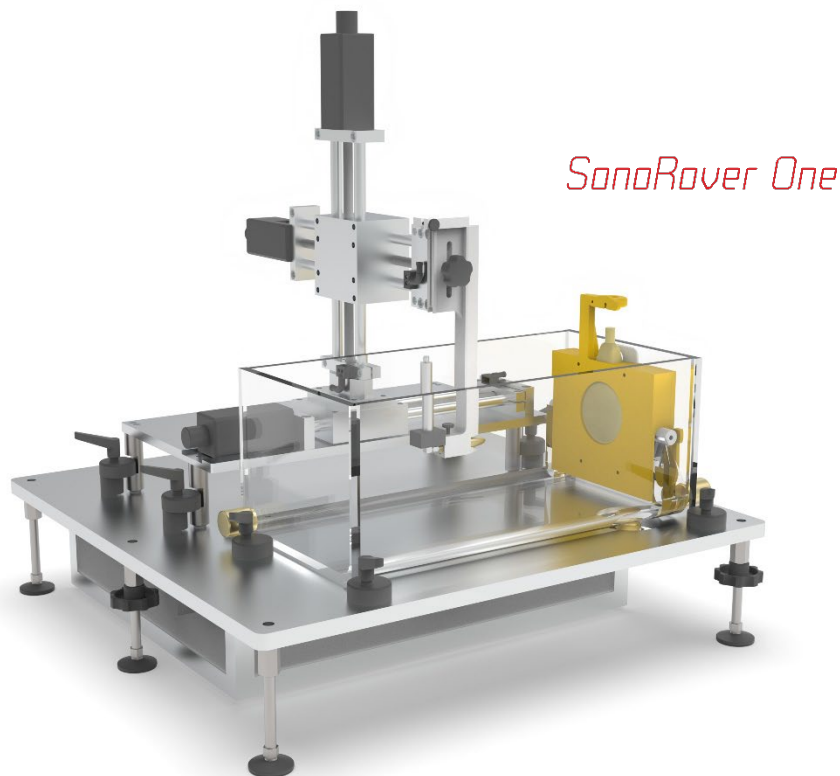


Setup manual

SonoRover One metrology platform



Document Identification and Filing

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Change log

Version	Date (yyyy-MM-DD)	Author(s)	Description of change(s)
0.7	2025-03-28	Stein Fekkes	Document structure.
0.8		Stein Fekkes	First release

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1 Introduction

1.1 Purpose & Scope

The SonoRover One Metrology Platform is an open-source hardware toolkit specifically designed for precise and efficient hydrophone measurements of low-intensity focused ultrasound (LIFU) systems. Its primary purpose is to provide researchers, developers, and clinicians with an accessible, reliable, and standardized method for characterizing acoustic performance across diverse ultrasound transducer portfolios.

The platform facilitates accurate acoustic metrology, enabling users to rapidly assess transducer characteristics such as acoustic intensity distribution, focal spot accuracy, and output consistency. By leveraging open-source acquisition systems, SonoRover One promotes transparency, reproducibility, and collaborative development within the focused ultrasound community.

The scope of this platform encompasses both research-oriented and clinical development environments, addressing the critical need for precision in acoustic measurement to ensure safety, efficacy, and standardization in therapeutic ultrasound applications.

This manual provides comprehensive guidance, including:

- Detailed explanations of the hardware system components and architecture
- Step-by-step instructions for initial setup and calibration
- Best practices for measurement accuracy and reproducibility

Through open collaboration, the SonoRover One Metrology Platform aims to foster innovation and accelerate advancements in focused ultrasound technologies by providing robust, standardized tools for acoustic characterization.

1.2 Audience

This manual is intended for professionals and researchers who are involved in the measurement, validation, and analysis of focused ultrasound systems. Typical users include:

- **Acoustic Engineers** working on transducer design, beam characterization, and system optimization.
- **Biomedical Researchers** involved in therapeutic ultrasound, neuromodulation, and preclinical validation of ultrasound-based devices.
- **Metrology and QA Specialists** responsible for compliance with acoustic safety guidelines and regulatory standards.
- **Lab Technicians and Operators** using the SonoRover One platform for routine measurements and reporting.
- **Graduate Students** or academic users working on ultrasound-related projects requiring hydrophone data analysis.

1.3 Environmental Requirements

To successfully setup and use the **SonoRover One Metrology Platform**, mechanical stability, Water condition, room conditions and safety and ergonomics should be taken into account. Make sure the supporting table is sturdy. The water remains stable at ambient temperature during the measurements in a climate controlled room with nominal humidity. No additional mechanical vibrations other than generated by the XYZ-stage.

1.4 Disclaimer

This user manual and associated documentation for the data analysis software tool provided herein are intended exclusively for guidance and informational purposes. Although every reasonable effort has been taken to ensure accuracy, correctness, and completeness, no guarantees or warranties—expressed or implied—are made regarding the software's functionality, reliability, suitability, or fitness for any particular purpose.

Users are solely responsible for verifying the accuracy and appropriateness of the software outputs and analyses, as well as the software's suitability for their specific data analysis tasks. The creators and distributors of the software disclaim all liability for any inaccuracies, errors, or consequences arising from the use, misuse, or interpretation of software results.

The user acknowledges that software performance may be influenced by factors including but not limited to system hardware configurations, operating environments, data integrity, and user input accuracy. Regular validation and verification procedures must be conducted by the user to ensure accurate and dependable results.

By using this software tool, the user agrees to assume full responsibility for compliance with applicable data protection, privacy regulations, and industry standards

2 Parts list

Refer to the CAD parts list stored at GIT (<https://github.com/Donders-Institute/Radboud-FUS-measurement-kit/tree/release/SonoRover%20One/hardware>) for a complete and detailed parts description. The list below is overview of the high level assemblies which might consist of subassemblies

- **XYZ stage**

Qty	Component	Description
1	3-Axis CNC frame*	CNC-style frame with linear rails and lead screws for hydrophone movement
	3 NEMA 17 Stepper Motors	
3	End stops and limit switches *	Used for safety range and homing procedure
1	Stage base plate	Stage base plate with 4 dowel pins for correct and reproducible alignment on parent base plate
1	Hydrophone bracket screw and washer*	To fixate the hydrophone bracket to the XYZ stage in a reproducible manner making use of a reference plate
1	Power adaptor	24V

- **Hydrophone**

Qty	Component	Description
1	Hydrophone bracket	3D SLA printed bracket minimizing direct acoustic reflections*
1	HGL Preamplifier	The HGL series of Onda are complemented with a cylindrical 90 degrees preamp AG-2010
1	HGL hydrophone	HGL 0200
1	Preamplifier offset block	Plastic block with a tight fitting around the preamp for correct and reproducible offset
1	BNC to preamplifier coax-cable	~60 cm long galvanic connection to DC power and oscilloscope connection
1	Power adaptor	24V 0.625A

- **Probe holders**

Qty	Component	Description
1	2-channel base holder	Grey holder which hosts all 2-channel NeuroFus transducers
1	4-channel base holder	Yellow holder which hosts all 4-channel NeuroFus transducers
1	Imasonic R100 and R75 base holder	Red holder which host R100 and R75 10 elements transducers
1	Back plate	The white back plate to push the transducer towards the front plate of the holder
1	2-channel probing cylinder	The centred hollow cylinder for probing exit plane
1	4-channel probing cylinder	The centred hollow cylinder for probing exit plane
1	Imasonic probing cylinder	The centred hollow cylinder for probing exit plane
1	Probing dummy hydrophone	The stainless steel dummy hydrophone

1	2-channel offset disk	Plastic offset disk for transducer fixation using the back plate
1	4-channel offset disk	Plastic offset disk for transducer fixation using the back plate
1	Imasonic offset disk	Plastic offset disk for transducer fixation using the back plate

- **Water tank**

Qty	Component	Description
1	Acrylic water tank	400x200x200 mm (LxWxH)
1	AptFlex F28 sheet	~200x200 mm High-frequency acoustic absorber designed to minimize unwanted ultrasonic reflections and provide effective acoustic isolation
2	In and output sockets	Brass quick connector for water hoses

- **Base plate**

Qty	Component	Description
1	Base plate	Aluminium baseplate 600x600 mm
3	Set feeds screws	Used to level the base plate
2	Lock feeds screws	If level, make sure this setting is kept
4	XYZ Fixation locks	XYZ stage lockers
1	Reference rig	The reference for positioning the water tank
3	Water tank fixation disk locks	Locking the water tank towards the reference rig and secure its position
4	Shock absorbers	Vibration absorber plates underneath the base plate feet's

- **Cabinet**

Qty	Component	Description
1	Cabinet	Box to contain all electronics and hovering above the floor
1	Oscilloscope	4 channel PicoScope with external power supply
1	Power strip	
1	Plug used for grounding	PicoScope grounds to net-ground
1	USB 3 hub	External powered USB 3 hub
1	50 Ohm BNC RF terminator	Impedance matching between the hydrophone and the oscilloscope

- **Pump and degasser**

Qty	Component	Description
1	pump	Degasser pump by IGT
1	Power adaptor	24V
2	Silicon hoses	1.5 meters of silicon hoses with quick -lock connectors in between pump and water tank

- **Peripherals**

Qty	Component	Description
-----	-----------	-------------

1	Dissolved oxygen and temperature measurement device	Hanna HI98198 portable dissolved oxygen measurement device
1	De-ionized water filter system	https://demiwatermaken.nl/product/saleskit/
2	Plastic containers 10L	For de-ionized water and waste water

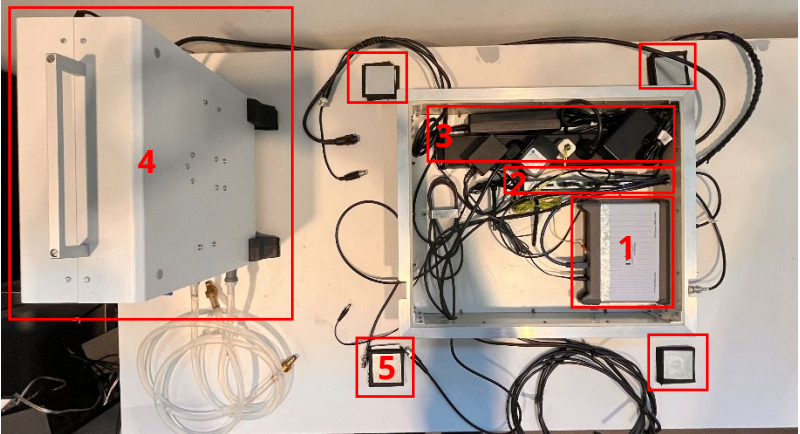
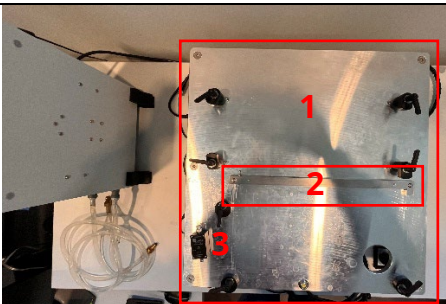
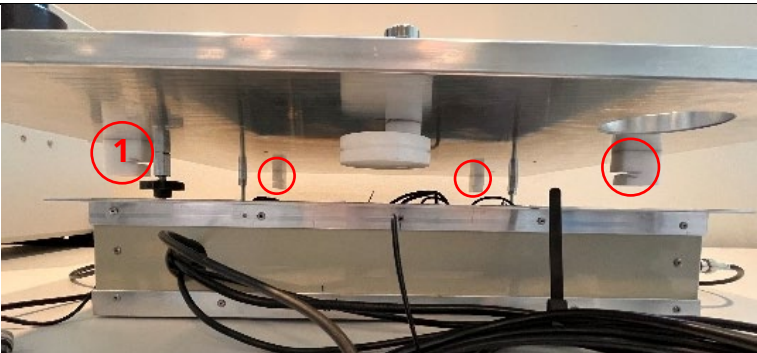
- Computer

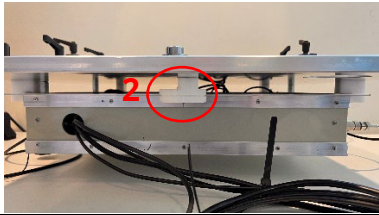
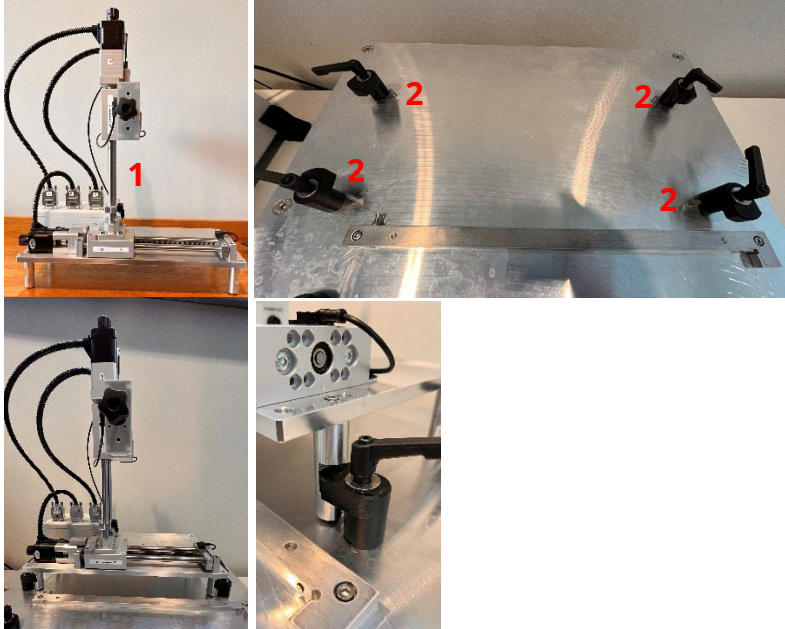
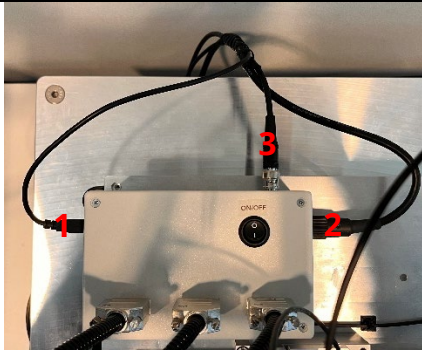
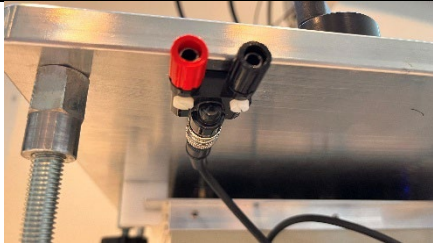
Qty	Component	Description
1	Desktop computer	Current desktop computer with fast USB-C data throughput
2	Desktop screens	
1	Keyboard	
1	Mouse	

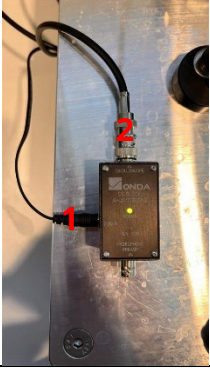
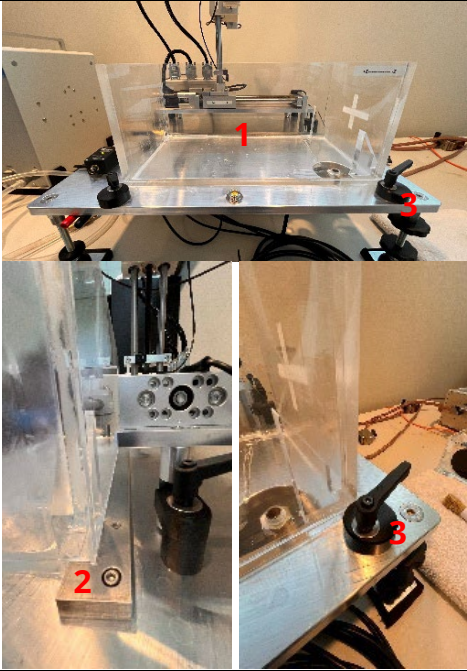
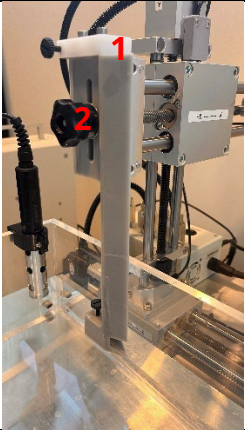
*Original by IGT and modified by Radboud University

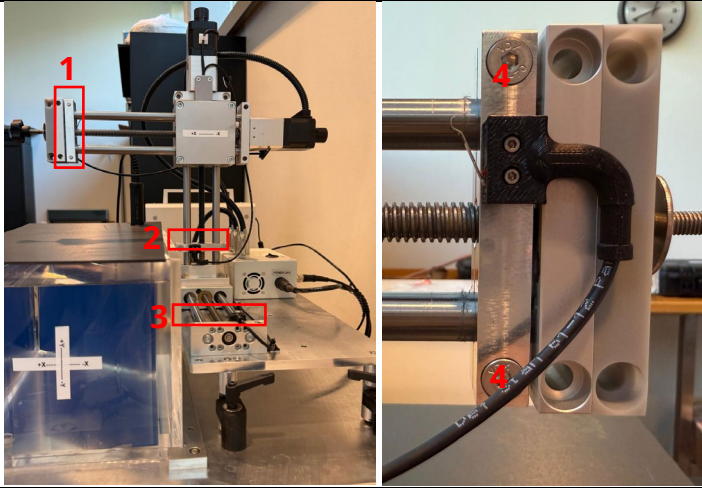
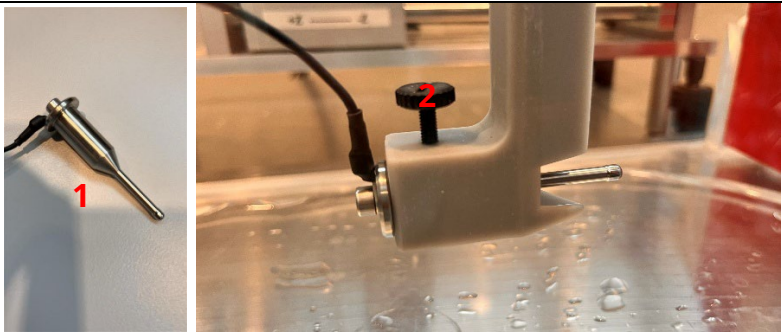
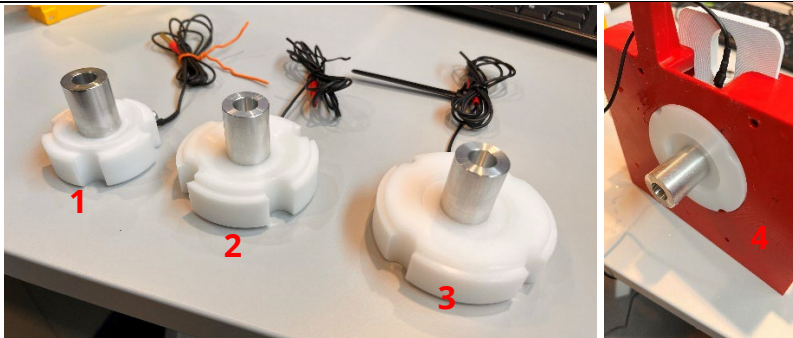
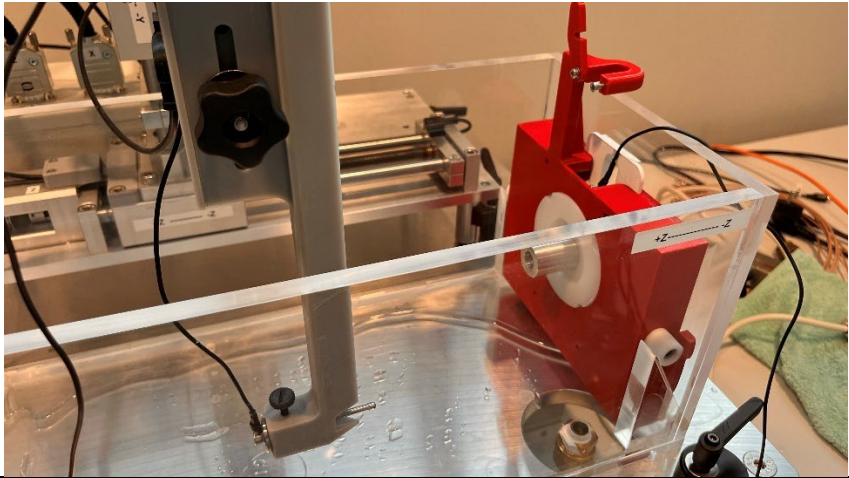
3 Installation

3.1 Initial setup instructions

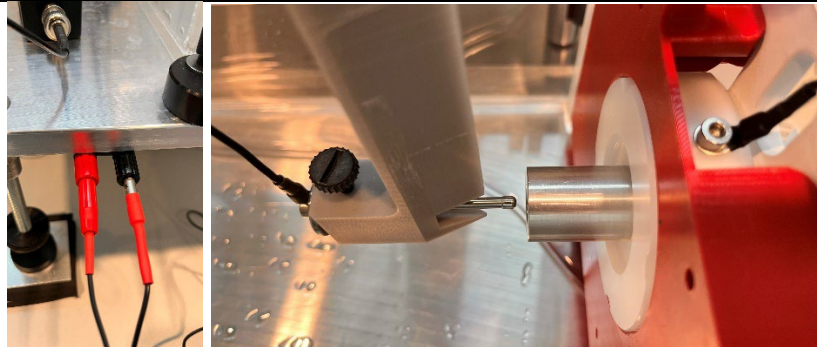
1.	Initial placement
Place the container box, pump and optional anti-vibration pads at a stable table 1) Oscilloscope 2) Powered USB 3 hub 3) Power socket 4) Pump/degasser 5) Anti-Vibration pads	
2.	
Position the base plate over the container box on top of the anti-vibration pads 1) Base plate 2) Reference rig 3) Hydrophone pre-amplifier	
3.	
Level the plate using the leveller on top by adjusting the screws at position 1,2,3... the anti-vibration pads will absorb minor changes 1) Lever screw 2) Locking screw 3) Leveller	
4.	
fixate the container below the base plate. Raise the whole container such that the sides are flush with the holders and push it towards the back. Secure it by rotation the front disk 1) Sliders 2) Locking disk	

		
5	Place the stage assembly at the baseplate over the cylinders and secure them with the fasteners 1) XYZ-stage 2) Stage lockers	
		
6	Connect the stage cables 1) Power 2) Data 3) BNC (probing)	
		
7	connect the BCN for mechanical alignment (probing) (the colours suggests polarity but this is not the case)	
		
8		

If applicable: connect the hydrophone preamplifier 1) Power 2) BNC to 50 ohm load prior to oscilloscope	
9 Place the water tank, and make sure its flush with the reference on the right upper corner. Then rotate the disks and fixate them with the bracket on top. To this for the upper left, lower left and lower right disks. This will secure the water tank in place and secure orthogonality w.r.t. the XYZ-stage orientation. 1) Water tank 2) Reference 3) Locking disks	
10 Mount the hydrophone connection bracket and fixate while pushing toward the left and downward against the references, then tighten the screw. 1) Reference 2) Locking screw	
11	

Mount the limit switch brackets on the X,Y and Z linear rails and adjust them such that the hydrophone cannot collide into the transducer surface or the sides of the water tank. 1) X-bracket 2) Y-bracket 3) Z-bracket 4) Hex screws for tighten the bracket	
Fixate dummy hydrophone by loosening the black screw first, insert the dummy hydrophone up to its reference and tighten with the screw 1) Dummy hydrophone 2) Locking screw	
12 Mount the dummy transducer in its appropriate probe holder and lock it into the water tank. 1) 2-channels NeuroFus Rig 2) 4-channels NeuroFus Rig 3) Imasonic R75/R100 Rig 4) Imasonic holder	 
13	

Turn on the XYZ stage switch. Connect both wires to the socket in front of the base plate. Run the "XYZ probing" sequence (see paragraph 3.3). Note the values and (need to be used before acoustical alignment). After the initial absolute coordinates are estimated, remove the dummy hydrophone and transducer from their respected brackets.



14

Connect the output pump side to the left input of the water tank and the input tank to a container with demi-water supply. Turn on the pump and degassing while filling up the tank. When filled completely, remove the input hose from the container and lock them to the bottom connector of the water tank.

- 1) Water tank input
- 2) Water tank output

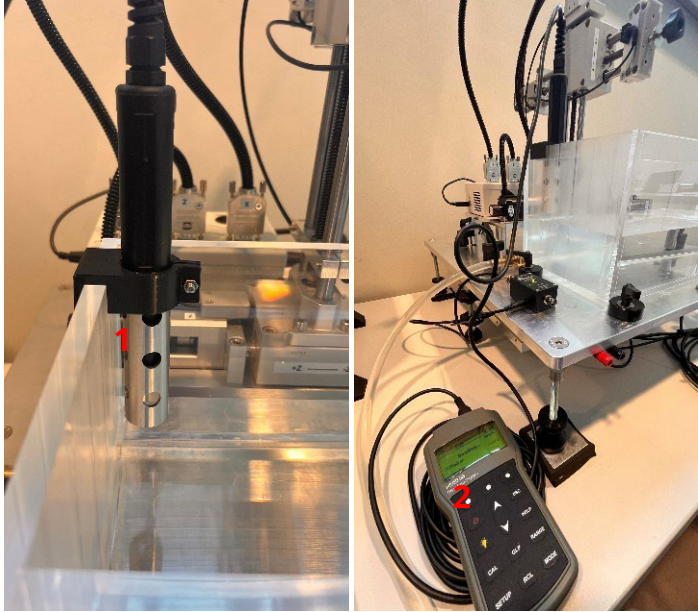
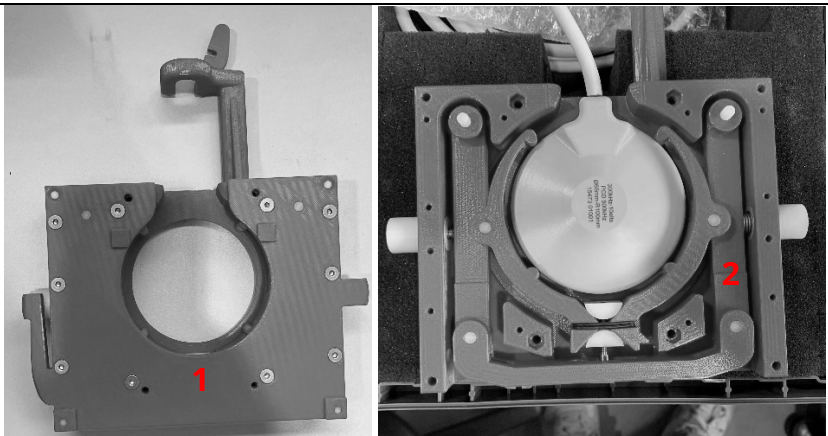


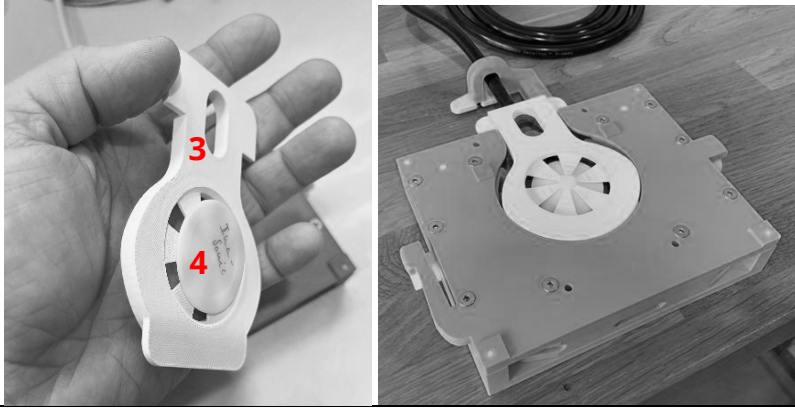
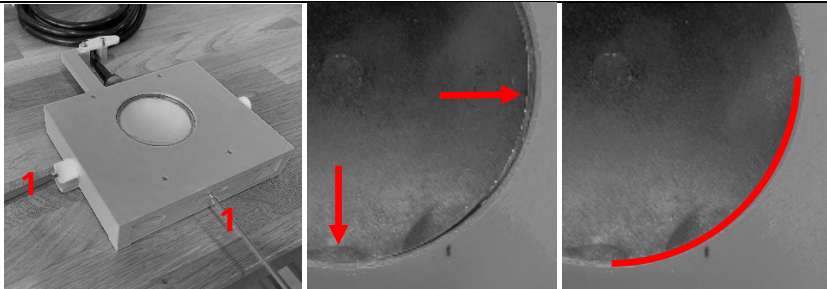
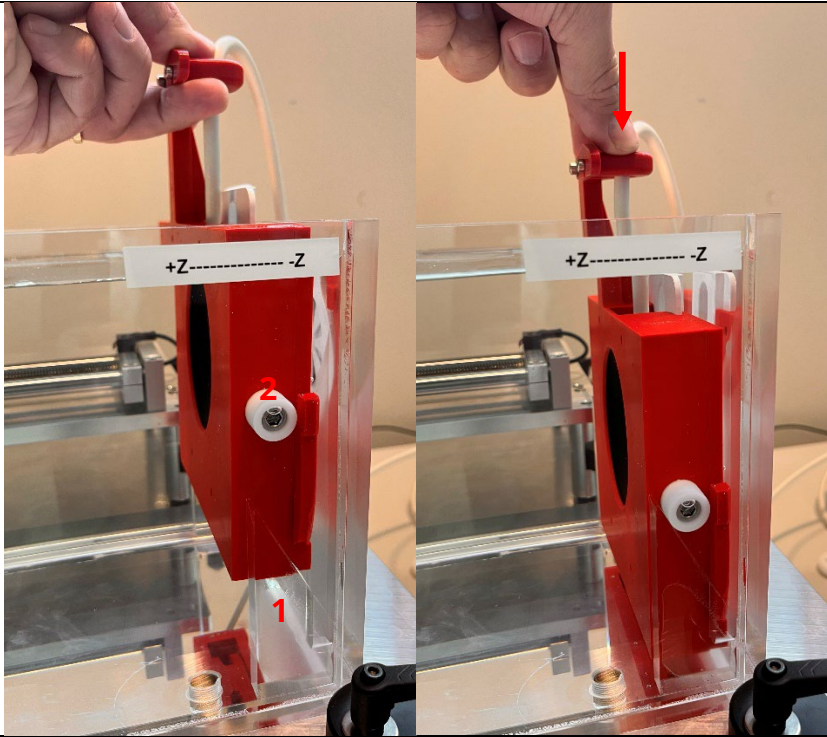
3.2 Pre-measurement setup instructions

After successfully completing the initial setup and calibration steps described in this manual, it's essential to perform specific preparation actions before each measurement session. These routine actions ensure optimal measurement accuracy, reproducibility, and reliability by maintaining consistent environmental conditions and system performance.

- **Check Environmental Conditions:**
Confirm water temperature, purity, and dissolved gas levels meet recommended criteria
- **Inspect Positioning System:**
Ensure all stage movements are functioning smoothly and verify the positioning accuracy.
- **System Warm-up and Stability:**
Allow sufficient warm-up time for all electronic measurement equipment, stabilizing electrical noise and drift. It is recommended to let the hydrophone acclimatize in the water for some time.

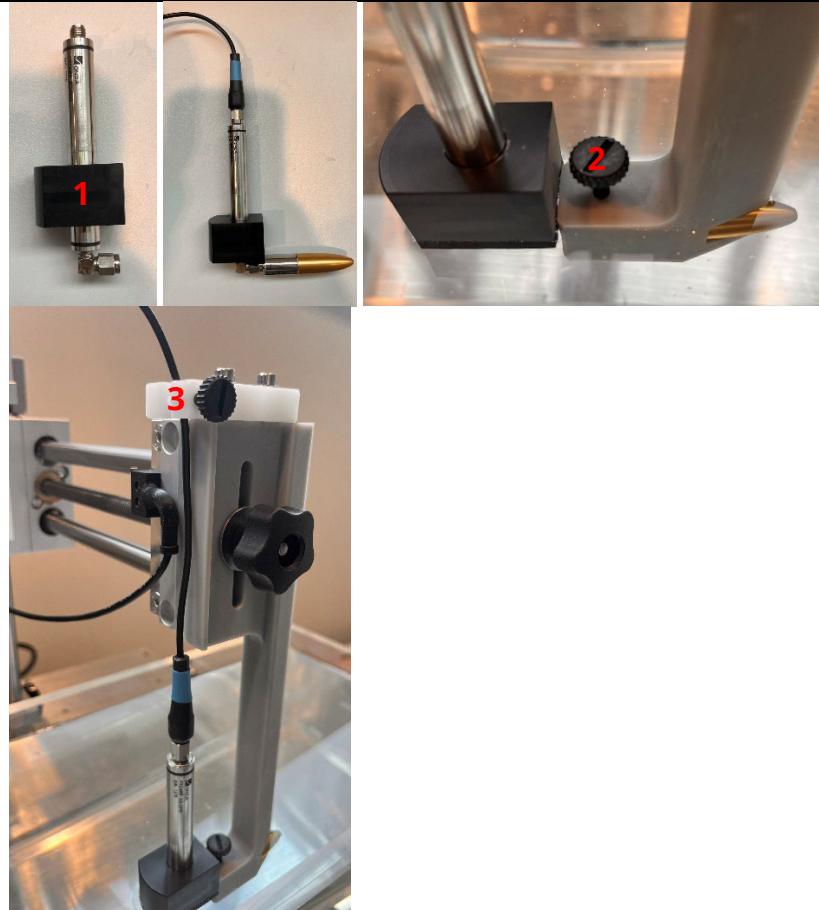
Performing these steps routinely before each measurement session guarantees precise and consistent results, significantly enhancing the overall reliability of your acoustic characterization process.

15 Place the dissolved oxygen measurement device in the water and start measuring the concentration oxygen. Remain pumping and degassing until the desired (<4mg/L) concentration has reached. Shut down the pump and repeat degassing upon undesired levels due to natural oxygenation of the water. 1) Dissolved oxygen and temperature probe 2) Portable readout	
16 Mount the transducer in the designated probe holder. 1) Probe holder 2) Inner body to manipulate probe center 3) Fixation bracket 4) Transducer specific offset ring	

	
17	
Align center 1) Inset appropriate hex tools at the bottom and side 2) Modify axial and lateral movement till probe is centered w.r.t. front opening	
18	
Mount probe holder in water tank and make sure that the probe holder is tightly locked by pushing it downwards firmly. This assures correct alignment to the back of the water tank and therefore to the XYZ-stage. 1) Water tank fixation wedges 2) Low friction cylinder knobs	
17	

Mount the hydrophone to the XYZ-stage. First slide the reference offset block over the pre-amplifier. Connect the hydrophone and the signal wire and slide till it abuts the holder and secure it with the screw. Guide the wire along the top of the hydrophone holder

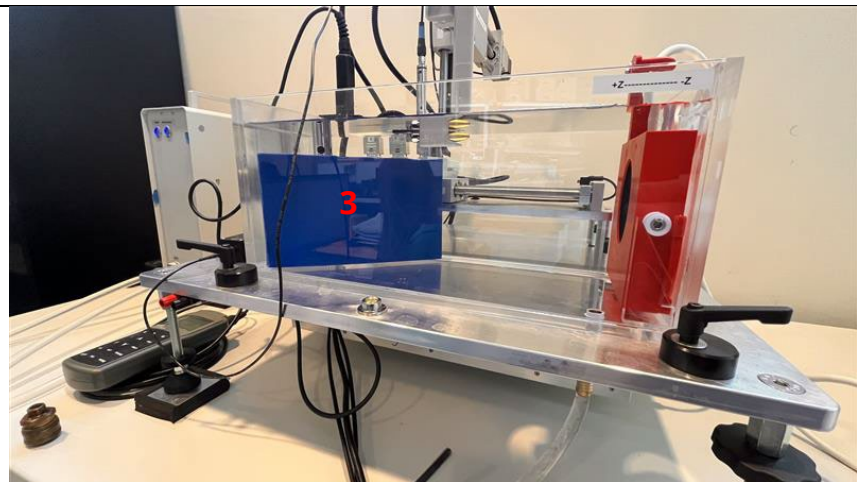
- 1) Reference offset block
- 2) Fixation screw
- 3) Wire guide



18

12 place acoustical reflection absorber sheet and fill with water

- 1) Acoustical anti-reflection sheet

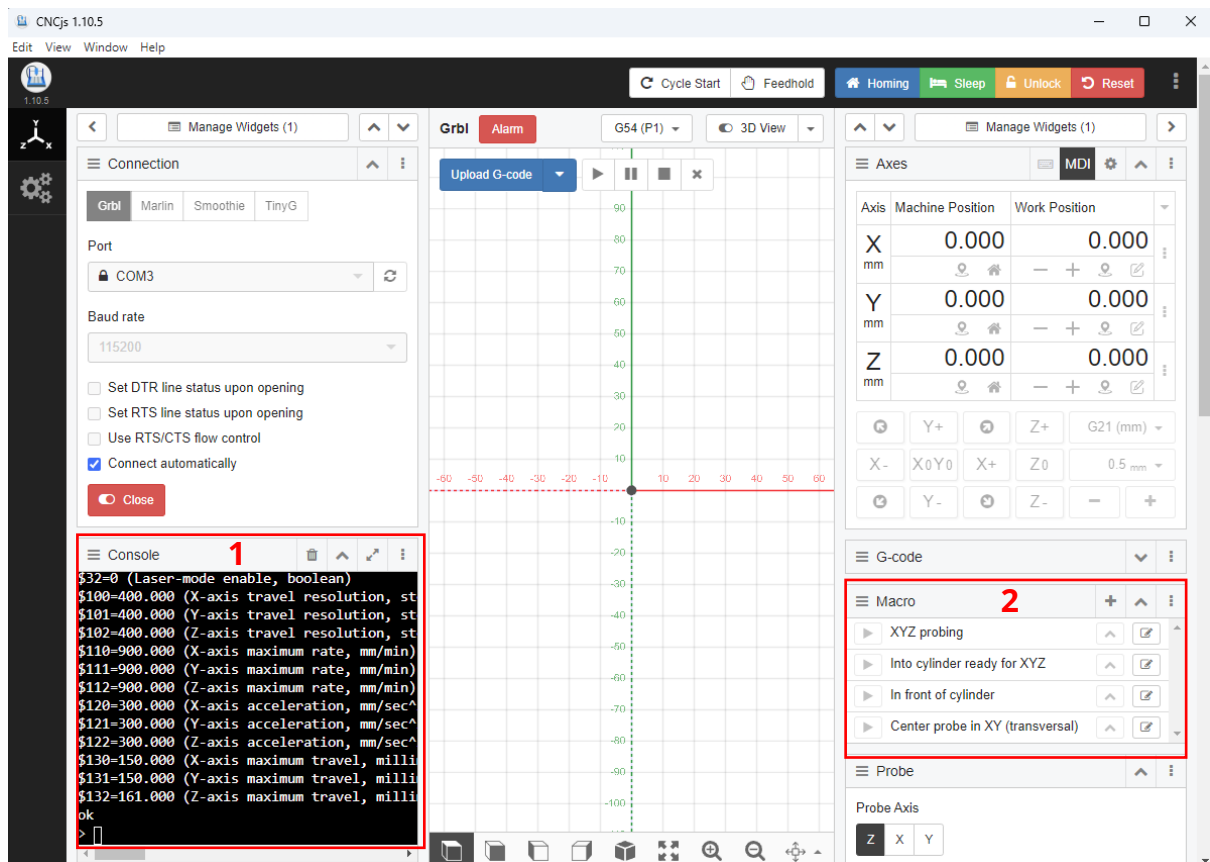


3.3 Software installation overview

- XYZ-Control system – CNCjs

CNCjs is an open-source, browser-based application used for precision control of motorized stages and positioning systems in three-dimensional space. It provides a user-friendly interface for accurately moving the hydrophone and ultrasound transducer, allowing precise alignment and repeatable measurements. CNCjs supports real-time manual controls, automated scripting, and live monitoring, making it ideal for precise metrology applications.

Download and Documentation: <https://cnc.js.org>. Make sure that the right com port is selected for successful serial communication.



CNCJS

- 1) Set the correct stage parameters with "\$" command to match:


```

Console
CNCjs 1.10.5 [Grbl]
Connected to COM3 with a baud rate of 115200
Grbl 1.1f ['$' for help]
client> $$
[MSG: '$H' '$X' to unlock]
$0=10 (Step pulse time, microseconds)
$1=25 (Step idle delay, milliseconds)
$2=0 (Step pulse invert, mask)
$3=0 (Step direction invert, mask)
$4=0 (Invert step enable pin, boolean)
$5=0 (Invert limit pins, boolean)
$6=0 (Invert probe pin, boolean)
$10=1 (Status report options, mask)
$11=0.010 (Junction deviation, millimeters)
$12=0.002 (Arc tolerance, millimeters)
$13=0 (Report in inches, boolean)
$20=0 (Soft limits enable, boolean)
$21=1 (Hard limits enable, boolean)
$22=1 (Homing cycle enable, boolean)
$23=0 (Homing direction invert, mask)
$24=50.000 (Homing locate feed rate, mm/min)
$25=600.000 (Homing search seek rate, mm/min)
$26=250 (Homing switch debounce delay, milliseconds)
$27=1.000 (Homing switch pull-off distance, millimeters)
$30=1000 (Maximum spindle speed, RPM)
$31=0 (Minimum spindle speed, RPM)
$32=0 (Laser-mode enable, boolean)
$100=400.000 (X-axis travel resolution, step/mm)
$101=400.000 (Y-axis travel resolution, step/mm)
$102=400.000 (Z-axis travel resolution, step/mm)
$110=900.000 (X-axis maximum rate, mm/min)
$111=900.000 (Y-axis maximum rate, mm/min)
$112=900.000 (Z-axis maximum rate, mm/min)
$120=300.000 (X-axis acceleration, mm/sec^2)
$121=300.000 (Y-axis acceleration, mm/sec^2)
$122=300.000 (Z-axis acceleration, mm/sec^2)
$130=150.000 (X-axis maximum travel, millimeters)
$131=150.000 (Y-axis maximum travel, millimeters)
$132=161.000 (Z-axis maximum travel, millimeters)
ok
>

```

- 2) Import a macro “XYZ probing” (see appendix) and make sure that the center of the acoustical axes is X-63.5 Y-62.0 w.r.t. the homing position determined by the end stops.

✕

Macro Name

XYZ probing

Macro Commands

+ Macro Variables ▾

; Thickness difference probe taster length - hydrophone length = 1,2 mm

; cylinder offset is 30.0 mm

; probe thickness 31.2 mm

\$h

G53 G90 G01 X-63.5 Y-62.0 F400

G53 G90 G01 Z-129 F400

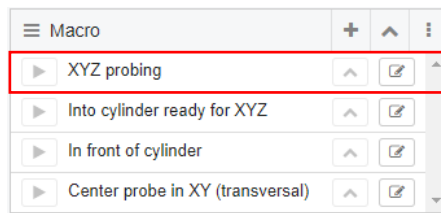
;Start with end mill in hole, BELOW Z surface of probe

Delete

Cancel

Save Changes

- 3)
- 4) Start the XYZ probing procedure to finally measure the X, Y and Z coordinate to use for the initial alignment procedure (see manual driving system software). In case of physical changes e.g., water tank replacement, please re-do this mechanical exit plane probing and set the correct initial values for the acoustical alignment. Please note that the initial mechanical alignment is the only input for the Z/axial coordinate. The X and Y are fine tuned using the axial alignment procedure ran before each measurement.



- Python IDE and manuals please refer to
 - <https://github.com/Donders-Institute/Radboud-FUS-measurement-kit/blob/release/README.md>
 - <https://github.com/Donders-Institute/Radboud-FUS-driving-system-software/blob/release/README.md>

4 Appendix

4.1 Macro for center exit plane probing

Please copy the below text into the Macro Commands field

Thickness difference probe taster length - hydrophone length = 1,2 mm
; cylinder offset is 30.0 mm
; probe thickness 31.2 mm

\$h

G53 G90 G01 X-63.5 Y-62.0 F400

G53 G90 G01 Z-129 F400

;Start with end mill in hole, BELOW Z surface of probe

; Wait until the planner queue is empty

%wait

; Set user-defined variables

%Z_PROBE_THICKNESS = 31.2 ;thickness of Z probe plate

%PROBE_DISTANCE = 10 ;Max distance for a probe motion

%PROBE_FEEDRATE_A = 100

%PROBE_FEEDRATE_B = 25

%PROBE_MAJOR_RETRACT = 1 ;distance of retract before probing opposite side

%Z_PROBE = 15 ; Lift out of hole and Max Z probe

%Z_PROBE_KEEPOUT = 2 ;distance (X&Y) from edge of hole for Z probe

%Z_FINAL = 5 ;final height above probe

%UNITS=modal.units

%DISTANCE=modal.distance

G91 ; Relative positioning

G21 ;Use millimeters

; Probe toward right side of hole with a maximum probe distance

G38.2 X[PROBE_DISTANCE] F[PROBE_FEEDRATE_A]

G0 X-2 ;retract 2mm

G38.2 X5 F[PROBE_FEEDRATE_B] ;Slow Probe

%X_RIGHT = posx

G0 X-[PROBE_MAJOR_RETRACT] ;retract

; Probe toward Left side of hole with a maximum probe distance

G38.2 X-[PROBE_DISTANCE] F[PROBE_FEEDRATE_A]

G0 X2 ;retract 2mm

G38.2 X-5 F[PROBE_FEEDRATE_B] ;Slow Probe

%X_LEFT = posx

%X_CHORD = X_RIGHT - X_LEFT

G0 X[X_CHORD/2]

```

%X_CENTER = posx      ;get X-value of hole center for some reason
; A dwell time of one second to make sure the planner queue is empty
G4 P1
G10L20X0

; Probe toward top side of hole with a maximum probe distance
G38.2 Y[PROBE_DISTANCE] F[PROBE_FEEDRATE_A]
G0 Y-2 ;retract 2mm
G38.2 Y5 F[PROBE_FEEDRATE_B] ;Slow Probe
%Y_TOP = posy
G0 Y-[PROBE_MAJOR_RETRACT] ;retract
; Probe toward bottom side of hole with a maximum probe distance
G38.2 Y-[PROBE_DISTANCE] F[PROBE_FEEDRATE_A]
G0 Y2 ;retract 2mm
G38.2 Y-5 F[PROBE_FEEDRATE_B] ;Slow Probe
%Y_BTM = posy
%Y_CHORD = Y_TOP - Y_BTM
%HOLE_RADIUS = Y_CHORD/2
G0 Y[HOLE_RADIUS]
%Y_CENTER = posy      ;get Y-value of hole center
; A dwell time of one second to make sure the planner queue is empty
G4 P1
G10L20Y0

;Get to Z probe location
G0 Z[Z_PROBE]
X[HOLE_RADIUS + Z_PROBE_KEEPOUT] Y[HOLE_RADIUS + Z_PROBE_KEEPOUT]

; Probe Z
G38.2 Z-[Z_PROBE] F[PROBE_FEEDRATE_A]
G0 Z2 ;retract 2mm
G38.2 Z-5 F[PROBE_FEEDRATE_B] ;Slow Probe
G10L20Z[Z_PROBE_THICKNESS]
G0 Z[Z_FINAL] ;raise Z
G90 ;absolute distance
G0 X0 Y0
; A dwell time of one second to make sure the planner queue is empty
G4 P1

[UNITS] [DISTANCE] ;restore unit and distance modal state
$h home dummy hydrophone

```


5 Changelog

5.1 Version history

March 2025	Initial version

6 Contact / Support

For direct contact: fus@ru.nl

6.1 Links to repository or project page (e.g., GitHub)

Resources (Solid works & Step files)	https://github.com/Donders-Institute/Radboud-FUS-measurement-kit/tree/release/SonoRover%20One/hardware
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